Naïve Bayes

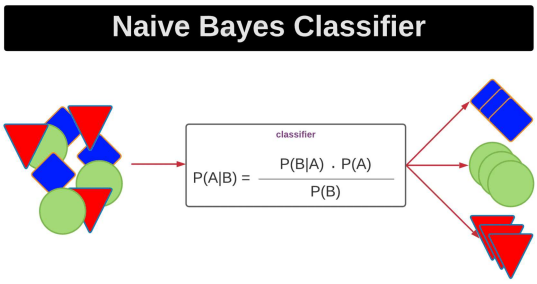
Monday, 27 October 2025 8:54 PM

Naive Bayes is like a detective who judges whether an email is "spam" or "ham" by looking for *clues* — words that have historically appeared more often in spam than in normal mail.

Each word is a witness.
Each class (spam or ham) is a courtroom.
The detective (our algorithm) listens to every witness, multiplies their credibility (likelihoods), and whichever courtroom has stronger overall testimony wins the case.

TOY EXAMPLE

Email	Text	Label
1	"buy cheap meds"	spam
2	"cheap meds available"	spam
3	"meeting schedule today"	not spam
4	"project meeting today"	not spam
5	"buy project plan"	not spam
6	"cheap meds project"	spam



Agenda → Use Case
→ Text Preprocessing (NLP)
→ Naive Bayes Algorithm - (Class -1)

(Google Gmail) → classifying emails as spam / Ham (Legitimate) (Not spam)

1. Mail 1 : "I, Nigerian prince need your help. Send money"
2. Mail 2: "Meeting scheduled for 5 pm. Kindly revert"

2 000(s) Email

Q. How did you come to conclusion that Mail 1 is Spam mail ?

2000

Q. What if we had basic approach that classifies spam from ham based on all these keywords?

→ (2 MB Hugging model) spam / Ham → 97% accuracy [No Machine Learning No DL]

(Analyse) → (EDA) → Are there certain words that occur frequently in spam email?,

(Analyse) → (EDA) → Are there certain words that occur frequently in spam email?

Preprocessing

2000(5)

	type	message
0	ham	Go until jurong point, crazy.. Available only ...
1	ham	Ok lar... Joking wif u oni...
2	spam	Free entry in 2 a wkly comp to win FA Cup fina...
3	ham	U dun say so early hor... U c already then say...
4	ham	Nah I don't think he goes to usf, he lives aro...

1. Tokenization

email = [see you today at 5]
3 20 10 7 6

[token = see, you, today, at, 5]

"Hi my name is Anthony Gonsalves." : ["my", 'Hi", "is", "Gonsalves", "Anthony", "name", "."]

2. Lower Casing

Q. Given two sentences, Should they be treated as different or same? → Lowercase

Sentence 1 : Let's go to the mall
Sentence 2: let's GO to the MALL

3. **. Removing punctuations** (. , ! , ? , ( ))

4). Remove Stop words

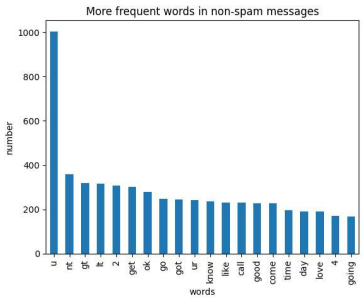
[a, an, the, do, it are]

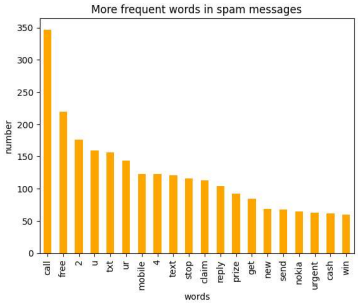
Q. Wouldn't removing stop words lead to grammatical errors? (It will)

Use Case Spam / Ham

→ nltk * Natural Language toolkit

→ spacy





want ?

$P(\text{spam}_1 \mid \text{email}) = ?$

$P(y=1 \mid \text{text}) = ?$

$P(y=0 \mid \text{text}) = ?$

Bayes Theorem ? (Revise)

A = it rains

B = ground is wet

Bayes Theorem

$$P(A \mid B) = \frac{P(A)\,P(B \mid A)}{P(B)}$$

$P(A \mid B)$: posterior prob

Since this is **conditional probability**, perhaps we can apply **Baye's theorm**?

Recall Baye's theorm from Probability/Stats lecture: **[PREREAD]**

- if A and B are some events then $P(A|B) = \frac{P(B|A) * P(A)}{P(B)}$

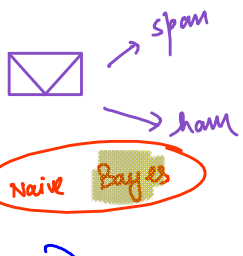
where,

P(A|B) = probability of A given B is true

P(B|A) = probability of B given A is true

P(A), P(B) = the independent probabilities of A and B

posterior



(1) $P(y=1 \mid \text{text}) = \frac{P(y=1)\,P(\text{text} \mid y=1)}{P(\text{text})}$

$P(y=1)$ or $P(\text{spam}) = \left\{ \frac{\#\ \text{spam emails}}{\#\ \text{Total email}} \right\}$

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(2) $P(y=0 \mid \text{text}) = \frac{P(y=0)\,P(\text{text} \mid y=0)}{P(\text{text})}$

$P(y=0)$ or $P(\text{ham}) = \frac{\#\ \text{ham}}{\#\ \text{Total}}$

Divide (1) and (2) → to cancel denominator

$$\frac{P(y=1 \mid x)}{P(y=0 \mid x)} = \frac{P(y=1) * P(x \mid y=1)}{P(y=0) * P(x \mid y=0)}$$

Zoom

$P(x \mid y=1) = P(\text{text} \mid \text{spam}) = P(\text{"my college is IISc Bangalore"} \mid \text{spam})$

$P(x|y=1) = P(\text{text}|\text{spam}) = P(\text{"my college is IISc Bangalore"}|\text{spam})$

Make an assumption (2000(s))

→ Naive Assumption

(Stupid) (Uninformed)



Naive

Naive Assumption → We know that this assumption do not hold true but still I am assuming

Quiz

Let's say:
the probability of dangerous fires are rare (1%)
but smoke is fairly common (10%) due to barbecues,
and 90% of dangerous fires make smoke
Can you find the probability of dangerous Fire when there is Smoke?

$P(F) = 1\% = 0.01$

$P(\text{smoke}) = 0.1 = 10\%$

$$P(F|S) = \frac{P(F)\,P(S|F)}{P(S)} = \frac{0.01 \times 0.9}{0.10} = \frac{0.9}{10} = 0.09$$

Let's say:
the probability of dangerous fires are rare (1%)
but smoke is fairly common (10%) due to barbecues,
and 90% of dangerous fires make smoke
Can you find the probability of dangerous Fire when there is Smoke?

3 options

Active Duration (Most preferred: 30 seconds)

Appears for 45 Secs

A 0.09

B 0.9

C 0.1

Break → 10:14 pm *

$P(x \mid y=1) = P(w_1, w_2, \ldots w_d \mid \text{spam})$

Exact seq is v.v.v. rare & hence we make a naive assumption → "Class Conditional Independence"

All features /words are conditionally independent given the class.

$P(w_1, w_2, \ldots w_d \mid \text{spam}) \xrightarrow{\text{If words are indep}} P(w_1 \mid \text{spam}) * P(w_2 \mid \text{spam}) * \ldots P(w_d \mid \text{spam})$

$P(\text{"my college is IISc Bangalore"} \mid \text{spam}) \xrightarrow[(C \cdot C \cdot I)]{\text{N.B assum}} P(\text{my} \mid \text{spam}) * P(\text{college} \mid \text{spam}) \times \ldots P(\text{Bangalore} \mid \text{spam})$

Why it's "naive" and when the assumption is false → all the time

- $P(w_1.w_2, w_3, \ldots, w_d \mid y=1) = P(w_1 \mid y=1) * P(w_2 \mid y=1) * \ldots * P(w_d \mid y=1)$

$P(w_1 \mid y=1) = \dfrac{\#(w_1) \text{ in all spam email}}{\#\text{ words in spam email}}$

Summary

$$\frac{P(y=1 \mid \text{text})}{P(y=0 \mid \text{text})} = \frac{P(y=1)\, P(w_1, \ldots w_d \mid \text{spam})}{P(y=0)\, P(w_1, \ldots w_d \mid \text{ham})}$$

$P(A, B) = P(A) * P(B)$

↓ Naive assumption (C·C·I)

$$P(y=1 \mid x) = \frac{P(y=1)}{P(y=0)} * \frac{P(w_1 \mid \text{spam}) * P(w_2 \mid \text{spam}) * \ldots P(w_d \mid \text{spam})}{P(w_1 \mid \text{ham}) * P(w_2 \mid \text{ham}) * \ldots P(w_d \mid \text{ham})}$$

$$\frac{P(y=1|x)}{P(y=0|x)} = \frac{P(y=1)}{P(y=0)} * \frac{P(w_1|ham) * P(w_2|ham) * \dots * P(w_{50}|ham)}{}$$

$\Rightarrow > 1 \Rightarrow$ (spam)

$< 1 \Rightarrow$ ham

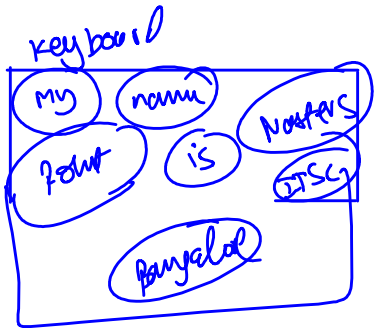
Naive Assumption → we assume "class conditional independence"

"once you know email is spam / not spam, the words are independent of each other"

$$P(w_1 \dots w_{150}|y) = P(w_1|y) \cdot P(w_2|y) \cdot \dots \cdot P(w_{150}|y)$$

Feel





why not just independence, why "**class conditioned**" independence

Naive Assumption states that,
- features are conditionally indepedent of each other
- i.e. for given class label =1, words are independent of each other
- similarly, for given class label = 0, words are independent of each other.

Which of the following statement is TRUE about the Bayes classifier?

optimal

4 options

Active Duration(Most preferred: 30 seconds)

Appears for	30 Secs

A	Bayes classifier is an unsupervised learning algorithm.
B	Bayes classifier works on the Bayes theorem of probability.
C	Bayes classifier is a Reinforcement Learning Algorithm
D	It assumes the independence between the independent variables or features.

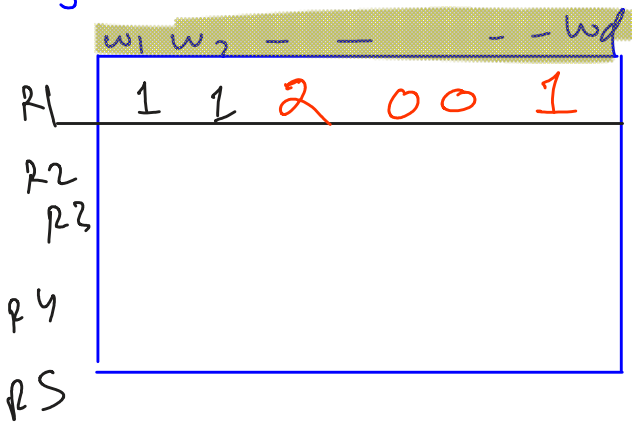
(Training Time)

Q. What will be features for Naive Bayes model ?

→ unique words in Traing data

$V = \{ w_1 w_2 - - \quad wd \}$

- "Nigerian prince needs help, send money"
- "You won a lottery, send money now"
- "Claim your million dollars"
- "Meeting scheduled for 5 pm, kindly reply"
- "Lunch tomorrow at 1 pm"



Training Time →

$P(y=1)$

$P(y=0)$

$P(w_j | y=1)$

$P(w_j | y=0)$

- we will have two constants, both the class priors
- $P(y=1) = \frac{Number\ of\ samples\ with\ y_i=1}{Total\ number\ of\ samples}$
- $P(y=0) = \frac{Number\ of\ samples\ with\ y_i=0}{Total\ number\ of\ samples}$
- And d likelihood values for both classes
- $P(w_j|y_i=1) = \frac{P(w_j \cap y_i=1)}{P(y_i=1)}$
- $P(w_j|y_i=0) = \frac{P(w_j \cap y_i=0)}{P(y_i=0)}$

What will be the Training time complexity of the Naive Bayes ?

Since we are to compute d likelihoods for all the n samples of the data

- the time complexity becomes **o(nd)**

Test Time

x_q New Email → "send money to Nigerian boy"

$$\frac{P(y=1|x_q)}{P(y=0|x_q)} = \frac{P(y=1)}{P(y=0)} * \frac{P(send|spam) * \cdots P(money|spam)}{P(send|ham) * \cdots P(buy|ham)}$$

$O(d)$

(Training Data)

Email text	y
"Nigerian prince needs help, send money"	1
"You won a lottery, send money now"	1
"Claim your million dollars"	1
"Meeting scheduled for 5 pm, kindly reply"	0
"Lunch tomorrow at 1 pm"	0

Test Time

E = "hurray" (w1), you (w2), won (w3), lottery (w4)

$$P(spam|E) = P(spam) * P(w_1|spam) * P(you|spam) * P(won|spam) * P(lott|spam)$$

* Since hurray was not in the Training data, the likelihood $= P(hurray|spam/ns) = 0 \Rightarrow P(spam|E) = 0$

odd small constant $= 10^{-6}$

Soln ⇒ Parameter fix 2

$$P(w_j \mid y=1) = \left[\frac{n_{j1} + \alpha}{n_1 + C\alpha}\right]$$

Assumption → (why are we studying)

- Extremely fast to train and predict.
- Works well with high-dimensional sparse data (text).
- Requires relatively little training data.
- Simple to implement and interpretable (you can inspect class-conditional probabilities).

Cons

- The independence assumption is ~~often~~ always false — can degrade performance if feature interactions are important.
- Output probabilities are often poorly calibrated; they should be used as scores rather than well-calibrated probabilities unless you calibrate them.
- Sensitive to how features are represented (counts, binary, normalized).
- Zero-frequency problem (handled by smoothing).